Siqi Fan

Institute for AI Industry Research (AIR), Tsinghua University

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Education

University of Chinese Academy of Sciences (UCAS) **Beijing, China**

- Master of Science in Automation *Sep. 2019 – Jun. 2022*
- School of Artificial Intelligence

Shanghai Jiao Tong University (SJTU) **Shanghai, China**

- Bachelor of Engineering in Automation *Sep. 2015 - Jun. 2019*
- School of Electronic Information and Electrical Engineering

Selected Publications

- **S. Fan**, F. Zhu, Z. Feng, et al. Conservative-progressive collaborative learning for semi-supervised semantic segmentation. In IEEE Transactions on Image Processing, (**IEEE TIP**), 2022.
- **S. Fan**, Q. Dong, F. Zhu, et al. SCF-Net: Learning spatial contextual features for large-scale point cloud segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, (**CVPR**), 2021.
- **S. Fan**, F. Zhu, S. Chen, et al. FII-CenterNet: An anchor-free detector with foreground attention for traffic object detection. In IEEE Transactions on Vehicular Technology, (**IEEE TVT**), 2021.
- **S. Fan**, F. Zhu, H. Zhang, et al. Improving road detection results based on ensemble learning and key samples focusing. In Proceedings of the IEEE International Conference on Intelligent Transportation Systems, (**ITSC**), 2020.

Research Experience

Institute for AI Industry Research, Tsinghua University (AIR) **Beijing, China**

Researcher, Computer Vision & Intelligent Vehicles *Jul. 2022 – Now*

- Works on multi-modal learning for scene understanding in complex environments
- Works on multi-view learning for cooperative perception of complex intelligent systems

Institute of Automation, Chinese Academy of Sciences (CASIA) **Beijing, China**

Student Researcher, Computer Vision & Intelligent Vehicles *Sep. 2019 – Jun. 2022*

- Works on environment perception for intelligent vehicles
  - **Introducing foreground attention to anchor-free detector for traffic object detection**
    - Proposed a foreground segmentation approach for anchor-free object detection, which could alleviate the background influences under complex traffic environment with little extra computation cost;
    - Generated the box-induced labels of foreground segmentation directly based on the bounding-box labels without additional labeling cost;
    - Proposed an anchor-free object detector with foreground attention;

- Achieved the advanced performance in both accuracy and efficiency.

- **Novel learning framework for semi-supervised semantic segmentation**
  - Proposed the conservative-progressive collaborating learning framework, which achieves the collaboration of conservative evolution and progressive exploration;
  - Proposed a pseudo labeling approach based on the class-wise confusion indicator for the disagreement part of paralleled branches;
  - Proposed a confidence-based dynamic loss to reduce the influences of noisy labels;
  - Achieved the SOTA performance, and the advantage is more evident when fewer labeled data is leveraged. The performance of full-supervision and few-supervision is improved.
- **Learning spatial contextual features for large scale point cloud segmentation**
  - Proposed a systematic approach for learning spatial contextual feature, including local information representation, local and global spatial contextual feature learning;
  - Designed the corresponding modules and a new 3D semantic segmentation network;
  - Achieved the SOTA performance for large scale point cloud semantic segmentation, and the performances on other related applications are verified by other researchers.

- Engineering projects: mapping and localization based on LiDAR, waypoints following based on multi-sensors, lane following based on camera, object detection based on the fusion of multi-modal information, data interchanging based on V2X.

Intel Labs China (ILC) **Beijing, China**

Autonomous System Research Intern, Intelligent Vehicles *Aug. 2020 – Dec. 2021*

- Worked on safety research for intelligent vehicles
  - **Simulation evaluation of Responsibility -Sensitive Safety (RSS)**
    - Evaluation of RSS on CARLA: build the test system, generate the test scenes and scenarios, conduct the experiments, analyze the experimental results, write the evaluation reports
    - Evaluation of RSS on MATLAB: refine the test system, conduct the experiments, analyze the experimental results, write the evaluation reports
  - **Field test of Responsibility -Sensitive Safety (RSS)**
    - Realized the test system: RTK-based perception, TCP-based transmission, CAN-based control
  - **Cross-team cooperation and academic communication with the partner universities**
  - **Participated in drafting the China ITS Industry Alliance Group Standard** "Technical Requirement of Safety Assurance of AV Decision Making"
  - The participated program was awarded **Intel's highest honor "Intel Achievement Awards"**

Intelligent Vehicle Lab, Shanghai Jiao Tong University **Shanghai, China**

Student Researcher, Computer Vision & Intelligent Vehicles *Jul. 2017 – Jun. 2019*

- Worked on object detection for intelligent vehicles
  - **LiDAR-based obstacle detection system on embedded device**
    - Initiated and led the "National Innovation Program for College Students".
  - **3D object detection via multi-modal fusion**

Honors and Awards

- **National Scholarship**, 2021
- **CASIA 'Pan Deng' First-class Scholarship**, 2022
- **Shanghai Jiao Tong University Excellent Scholarship**, 2018
- **China Industrial Intelligence Challenge, Outstanding Award (State Level)**, 2018

Academic Services

- **Invited talks and presentations**
  - Traffic scenes understanding and simulation testing, Invited talk at IEEE International Conference on Intelligent Transportation Systems 2022 workshop, September 2022, Online.
- **Reviewer of**

  IEEE Transactions on Image Processing (**IEEE TIP**), IEEE Transactions on Circuits and Systems for Video Technology (**IEEE TCSVT**), IEEE Transactions on Vehicular Technology (**IEEE TVT**), IEEE Transactions on Intelligent Vehicles (**IEEE TIV**), Pattern Recognition (**PR**), IET Computer Vision (**IET CV**), IET Cyber-Systems and Robotics (**IET CSR**), IEEE Intelligent Transportation Systems Magazine (**IEEE ITSM**), IEEE International Conference on Intelligent Transportation Systems (**IEEE ITSC2022**)